

Global Temperature, Atmospheric CO₂, and Solar Variability: A Reproducible Time-Series Analysis

A reproducible time-series analysis of the NASA GISTEMP temperature record, the NOAA Mauna Loa CO₂ record, and the SILSO sunspot record.

Abstract. We analyse three real, publicly available climate time series: 147 years of global land–ocean temperature anomaly (NASA GISTEMP), 69 years of Mauna Loa CO₂ (NOAA GML / Keeling curve), and 278 years of monthly sunspot counts (SILSO). Using robust trend estimation with significance testing, breakpoint analysis, seasonal–trend decomposition (STL), Fourier spectral analysis, partial correlation, and a three-model forecasting comparison, we obtain the following results. (1) Global temperature rose at 0.082 °C per decade over the full record (95% CI [+0.073, +0.091]; Mann–Kendall $p = 1.41\text{e-}39$ under independent observations; autocorrelation-robust check in §5.2), with statistically significant acceleration after 1970 (post-1970 rate +0.204 °C per decade). (2) CO₂ rose at 16.422 ppm per decade (CI [+15.724, +17.155]; $p = 1.17\text{e-}98$), and the amplitude of its seasonal cycle increased by +0.146 ppm per decade (CI [+0.116, +0.179]; $p = 8.86\text{e-}12$). (3) Spectral analysis recovers a dominant solar period of 11.10 years. (4) The annual temperature–CO₂ association is strong ($r = +0.97$) and persists after controlling for the solar cycle (partial $r = +0.97$; residual $r^2 = 0.94$), whereas the residual solar association is small (partial $r = +0.35$; $r^2 = 0.12$) and not robust to first-differencing. (5) On a 127-month holdout (2016-01 to 2026-07), a seasonal ARIMA model outperforms both XGBoost and a small LSTM on both series; all three methods beat a persistence baseline for CO₂, but only ARIMA does so for temperature. All claims are traceable to the scripts listed in Section 8.

Glossary of abbreviations

Abbreviation	Full form
NASA	National Aeronautics and Space Administration (US)
GISTEMP	GISS Surface Temperature Analysis — the NASA global temperature record
CO ₂	Carbon dioxide
NOAA	National Oceanic and Atmospheric Administration (US)
GML	Global Monitoring Laboratory (NOAA)
SILSO	Sunspot Index and Long-term Solar Observations (World Data Center, Brussels)
WDC	World Data Center
SSN	Sunspot number
STL	Seasonal-Trend decomposition using Loess
Loess	Locally estimated scatterplot smoothing
FFT	Fast Fourier Transform
ARIMA	Autoregressive Integrated Moving Average
SARIMA	Seasonal ARIMA
AIC	Akaike information criterion
XGBoost	eXtreme Gradient Boosting
LSTM	Long Short-Term Memory (a type of recurrent neural network)
MAE	Mean absolute error
RMSE	Root mean squared error
CI	Confidence interval
SD	Standard deviation
ppm	Parts per million
p	Probability value (statistical significance)
r	Pearson correlation coefficient
τ (tau)	Kendall's rank correlation coefficient
z	Standard score (units of standard deviation)

1. Introduction

Climate change is arguably the most consequential measurement program in the history of the physical sciences, and its evidentiary core consists of a small number of long, carefully maintained time series. Three of these records stand out. The global land–ocean temperature anomaly, compiled by NASA GISTEMP since 1880, is the most direct measure of whether the planet is warming. The Mauna Loa CO₂ record — the famous "Keeling curve" begun by Charles Keeling in 1958 — documents the principal greenhouse gas associated with that warming. And the sunspot number, compiled by the World Data Center SILSO since 1749, tracks the natural solar variability that is the leading candidate alternative explanation. Together, the three records provide a basis for examining warming trends, greenhouse-gas concentrations, and solar variability. All three are real, publicly available measurements, downloaded once and never modified by this analysis.

Because these records are measurements through time, the scientific questions about them are time-series questions. Is there a trend, and is it statistically significant? Does the rate of change itself change? What periodic structure does a series contain, and how has it evolved? Which external series best explains the variation of another, once other influences are removed? And can the series be forecast, and by which method? The toolkit of time-series analysis — robust trend estimation, significance testing, seasonal decomposition, spectral analysis, partial correlation, and forecasting — was developed precisely for such questions, and each tool is exercised in this study.

The study is organised as a reproducible pipeline of analytical stages, each implemented as a single runnable script that reads the raw data and produces its own figures and numbers (Section 8). Trends are never reported without their significance; the breakpoint follows the project plan; observed results are kept separate from their interpretation; and the forecasting comparison is judged exclusively on data not seen during fitting. This report presents each stage's figure and table with a plain-English reading of what they show.

The study addresses three questions:

1. **Is the planet warming, and how is this established?** The temperature trend is estimated with a robust Theil–Sen slope and tested with the Mann–Kendall test, so the answer carries a significance level rather than resting on visual inspection; a breakpoint analysis then asks whether the warming rate has itself changed since 1970.
2. **Why does atmospheric CO₂ rise and oscillate every year?** The Keeling curve is decomposed into a secular trend, a seasonal cycle, and residual noise, and the amplitude of the seasonal cycle is measured over time — asking not only what the curve does, but whether its behaviour is changing.
3. **How much of recent warming is solar?** The temperature record is compared with the sunspot cycle and with CO₂ using partial correlations, separating the two candidate drivers; the records are then used in a three-model forecasting comparison that asks which method —

classical statistics, tree-based machine learning, or a neural network — best predicts the near future on an unseen holdout.

Answering these questions exercises the full time-series toolkit on data that matter: significance-tested trends, structural change, decomposition, periodicity, association, and forecasting, with every claim traceable to a reproducible step.

2. Data and provenance

Table 1. Datasets used in this study.

Dataset	Source	Coverage	Scientific role
Global land–ocean temperature anomaly (monthly, °C vs 1951–80)	NASA GISTEMP v4	1880–2026	Primary target series
Mauna Loa CO2 (monthly, ppm)	NOAA GML (Keeling curve)	1958–2026	Greenhouse-gas driver; trend and seasonality
Sunspot number (monthly SSN)	SILSO, WDC-SILSO Brussels	1749–2026	Natural solar variability; control variable

Raw files are downloaded once to `data/raw/` and are never modified by analysis code. GISTEMP reports the most recent months (from 2026-08) as missing; these are excluded wherever they would affect a calculation.

3. Methods

Trend estimation and significance testing follow the standard climatological toolkit. The trend in each series is estimated with the Theil–Sen estimator — the median of the slopes of all pairs of observations — which resists the influence of outliers far better than ordinary least squares. Its significance is assessed with the Mann–Kendall test, the standard non-parametric test for monotonic trend; the associated p-value quantifies the probability of observing the pattern under a null hypothesis of no trend. The p-values assume independent observations; because climate series are autocorrelated, Section 5.2 reports a robustness check with an effective sample-size correction. Every trend estimate in Section 5 is reported with its 95% confidence interval and p-value, so that each claim carries its significance rather than a bare slope.

The structure of the series is characterised with two further tools. STL decomposition splits a monthly series into trend, seasonal, and residual components by locally weighted smoothing (Loess), allowing each component to be analysed separately; it is applied to the CO2 record to separate the secular rise from the annual cycle and to measure the cycle's amplitude over time. Spectral analysis uses the Fourier periodogram — the power of the series as a function of period — to identify dominant cycles; a

Hann window and linear detrending are applied before the transform so that the trend does not mask the periodic signal, and the tool is used to recover the solar cycle from the sunspot series.

Questions about statistical association are addressed with partial correlation, which measures the association between two variables after the linear effect of a third has been removed from both; this allows the analysis to ask whether temperature tracks CO₂ once the solar cycle is accounted for, and vice versa. Finally, the forecasting comparison follows a strict protocol: every model is fitted once on data up to December 2015 and evaluated exclusively on the 127-month holdout (2016-01 to 2026-07), with MAE and RMSE computed against a persistence baseline. The models and their configuration are described in Section 4.

4. Forecasting models and configuration

Table 2. Forecasting models: configuration and rationale (fixed in advance; nothing tuned on the test period).

Model	Configuration	Rationale
SARIMA	statsmodels ARIMA, d=1, seasonal (0,1,1,12) (airline model); p,q in {0,1,2} chosen by AIC on the training set	Monthly data require a seasonal term; the airline model is the standard parsimonious fit for strongly seasonal series such as the Keeling curve; AIC gives a principled, data-driven choice of the AR/MA orders
XGBoost	Gradient-boosted trees; 12 monthly lagged differences as features; 200 trees, max depth 4, learning rate 0.05; random_state 42; forecasts built recursively	Differenced targets prevent the drift that recursive forecasting on levels exhibits; 12 lags capture the annual cycle; the configuration is a standard mid-size boosting setting
LSTM	Recurrent network; 12 lagged differences (z-scored) fed as a 12-step sequence; one hidden layer of 16 units; linear output head; MSE loss; Adam lr 1e-3; batch 64; 30 epochs; seed 42	Tests whether a neural model that learns its own features can beat engineered baselines; z-scoring stabilises training; the small architecture suits 700-1600 training points
Persistence	Forecast equals the last observed value	The naive floor: any model that cannot beat it adds no information

Three modelling families are compared: a classical statistical model (SARIMA), a tree-based machine-learning model (XGBoost), and a neural network (LSTM), with persistence as the baseline (Table 2). The trio spans the methodological spectrum and is the standard comparison set for time-series forecasting. Two configuration choices are shared by the machine-learning models and deserve emphasis. First, the prediction target is the monthly *difference* rather than the level; recursive forecasting on levels of a trending series compounds small biases into large drift, and differencing removes this failure mode (it reduced XGBoost's temperature MAE from 1.01 to 0.34 °C in

development). Second, the 12-month lag window matches the annual cycle, giving the models the information needed to reproduce seasonality without imposing its functional form. All hyperparameters are fixed before the test period is examined; the comparison therefore reflects the models as configured, not after optimistic selection.

5. Results

5.1 Overview of the data

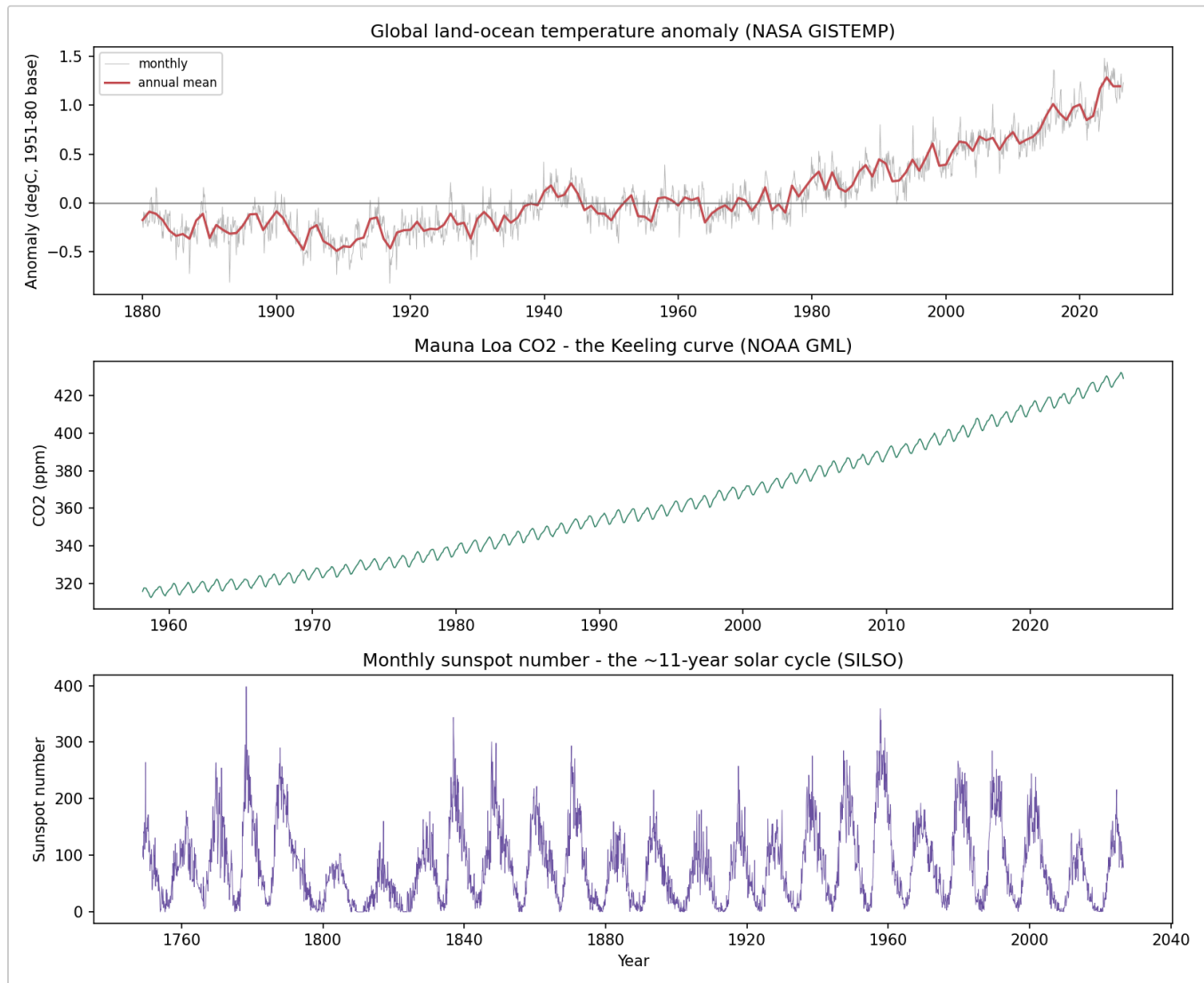


Figure 1. Overview of the three records: monthly temperature anomaly with annual means (inset red), the Mauna Loa CO2 record, and monthly sunspot numbers.

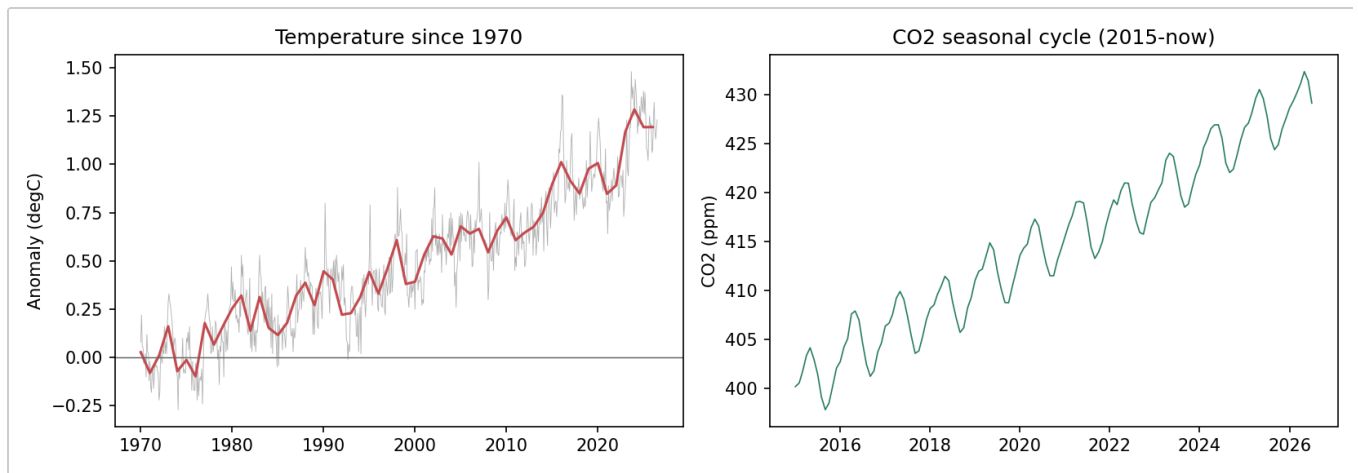


Figure 2. Detail: temperature anomaly since 1970, and the CO2 seasonal cycle since 2015.

Table 3. Summary statistics for the three series (monthly observations).

Series	n	Start	End	Latest	Mean	SD	Min	Max
GISTEMP temp anomaly (degC)	1759	1880-01-01	2026-07-01	1.23	0.09	0.42	-0.82	1.48
Mauna Loa CO2 (ppm)	821	1958-03-01	2026-07-01	429.12	361.28	33.40	312.42	432.34
Sunspot number (SSN)	3331	1749-01-01	2026-07-01	78.10	82.20	67.58	0.00	398.20

Table 4. Decadal mean temperature anomalies (°C, 1951–80 base).

Decade	Mean anomaly
1880	-0.21
1890	-0.24
1900	-0.32
1910	-0.33
1920	-0.24
1930	-0.12
1940	0.04
1950	-0.05
1960	-0.03
1970	0.03
1980	0.25
1990	0.38
2000	0.59
2010	0.81
2020	1.08

The monthly temperature series exhibits substantial short-term variability, yet the annual means display a clear secular increase from approximately $-0.2\text{ }^{\circ}\text{C}$ in the 1880s to $+1.2\text{ }^{\circ}\text{C}$ in the 2020s (Figure 1, upper panel). The CO₂ record combines a monotonic upward trajectory with a regular annual oscillation — the seasonal exchange of carbon between the atmosphere and the terrestrial biosphere (Figure 2, right panel). The sunspot series shows a persistent quasi-periodic modulation on a decadal timescale, consistent with the well-documented solar activity cycle (Figure 1, lower panel). Decadal temperature means (Table 4) change from $-0.21\text{ }^{\circ}\text{C}$ in the 1880s to $+1.08\text{ }^{\circ}\text{C}$ in the 2020s, a cumulative increase of $+1.29\text{ }^{\circ}\text{C}$ that foreshadows the formal trend analysis of Section 4.2.

5.2 Trend analysis

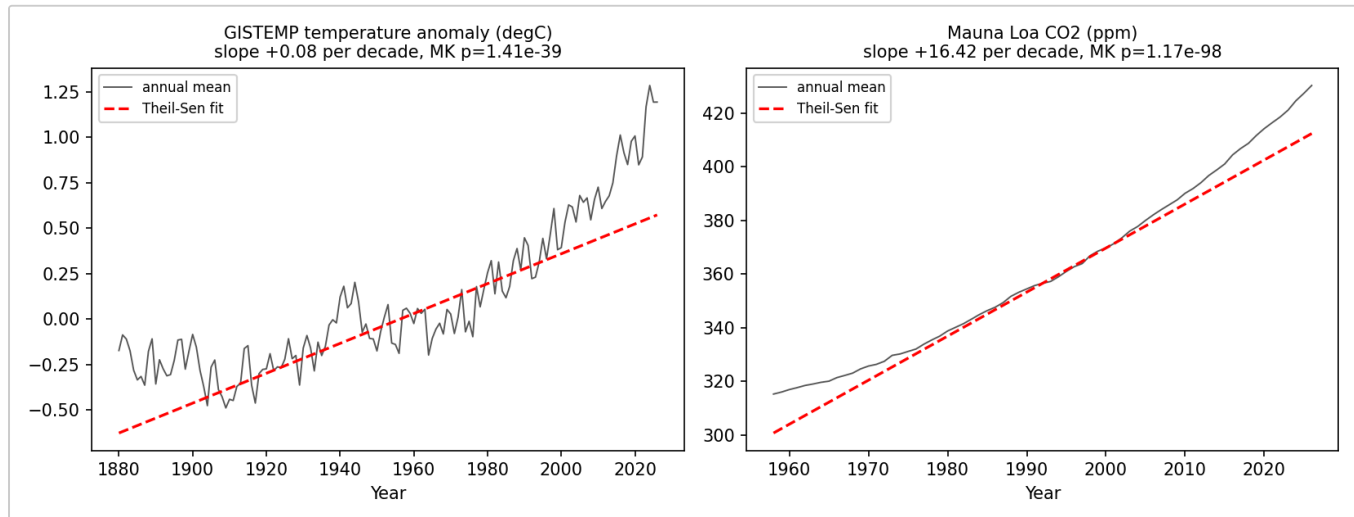


Figure 3. Annual means with fitted Theil–Sen trend lines for temperature (left) and CO₂ (right).

Table 5. Theil–Sen slopes on annual means, with 95% confidence intervals and Mann–Kendall statistics; p-values assume independent observations (robustness check in §5.2).

Series	n (years)	Slope (per decade)	95% CI	Tau	p-value
GISTEMP temperature anomaly (°C)	147	0.082	[+0.073, +0.091]	0.733	1.41e- 39
Mauna Loa CO ₂ (ppm)	69	16.422	[+15.724, +17.155]	1.000	1.17e- 98

Table 5 reports the trend estimates and their significance. Temperature rose at 0.082 °C per decade over 1880–2026 (95% CI [+0.073, +0.091]; $p = 1.41\text{e-}39$); the probability of such a pattern under a null hypothesis of no trend is on the order of 10^{-39} , so the warming signal cannot plausibly be attributed to sampling variability. CO₂ rose at 16.422 ppm per decade (CI [+15.724, +17.155]; $p = 1.17\text{e-}98$), equally unambiguous. In both panels of Figure 3 the fitted lines track the annual means closely, and because the Theil–Sen estimator is a median of pairwise slopes, the estimates are robust to the influence of individual extreme years.

Supplementary robustness check (autocorrelation). The Mann–Kendall p-values above assume independent annual observations, which understates the uncertainty for autocorrelated climate series. Recomputing the test with an AR(1)-based effective sample size (a modified Mann–Kendall correction) leaves the temperature trend significant at $p \approx 8 \times 10^{-5}$ (effective $n \approx 13$ of 147 years) and the post-1970 trend at $p \approx 4 \times 10^{-11}$ (effective $n \approx 29$ of 57). For CO₂ the detrended series is so strongly autocorrelated (lag-1 $\rho \approx 0.99$) that the correction collapses the effective sample size and the corrected p-value is unreliable; the trend statistic itself is unchanged (Kendall's $\tau = 1.00$, a perfectly monotonic series). The substantive conclusions of this section are unaffected by the correction.

5.3 Breakpoint analysis: acceleration after 1970

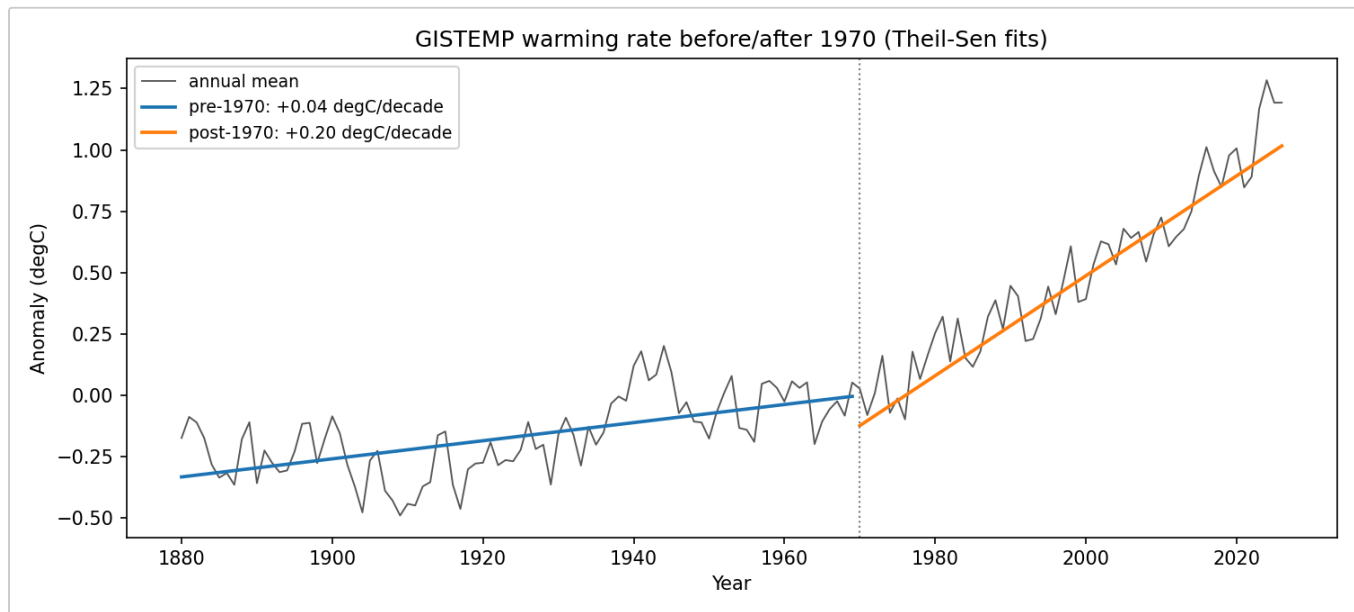


Figure 4. Theil–Sen fits applied separately to the pre-1970 and post-1970 segments of the temperature record.

Table 6. Warming rates for the two segments and the full record; p-values assume independent observations (robustness check in §5.2).

Segment	n (years)	Slope (°C/decade)	95% CI	p-value
pre-1970	90	+0.037	[+0.026, +0.048]	5.06e-09
post-1970	57	+0.204	[+0.182, +0.222]	4.17e-20

The warming rate increased from +0.037 °C per decade before 1970 to +0.204 °C per decade after 1970 (Table 6). Because the 95% confidence intervals of the two estimates do not overlap, the acceleration — +0.167 °C per decade per decade — is statistically significant. The change in gradient at the break is readily apparent in Figure 4. The breakpoint (1970) follows the project plan (docs/process.md), which fixes the research question as the warming rate before and after ~1970; a single split was examined and no alternative candidate years were scanned, so the significance statement is not inflated by data-dependent selection.

5.4 Seasonal decomposition of the CO2 record

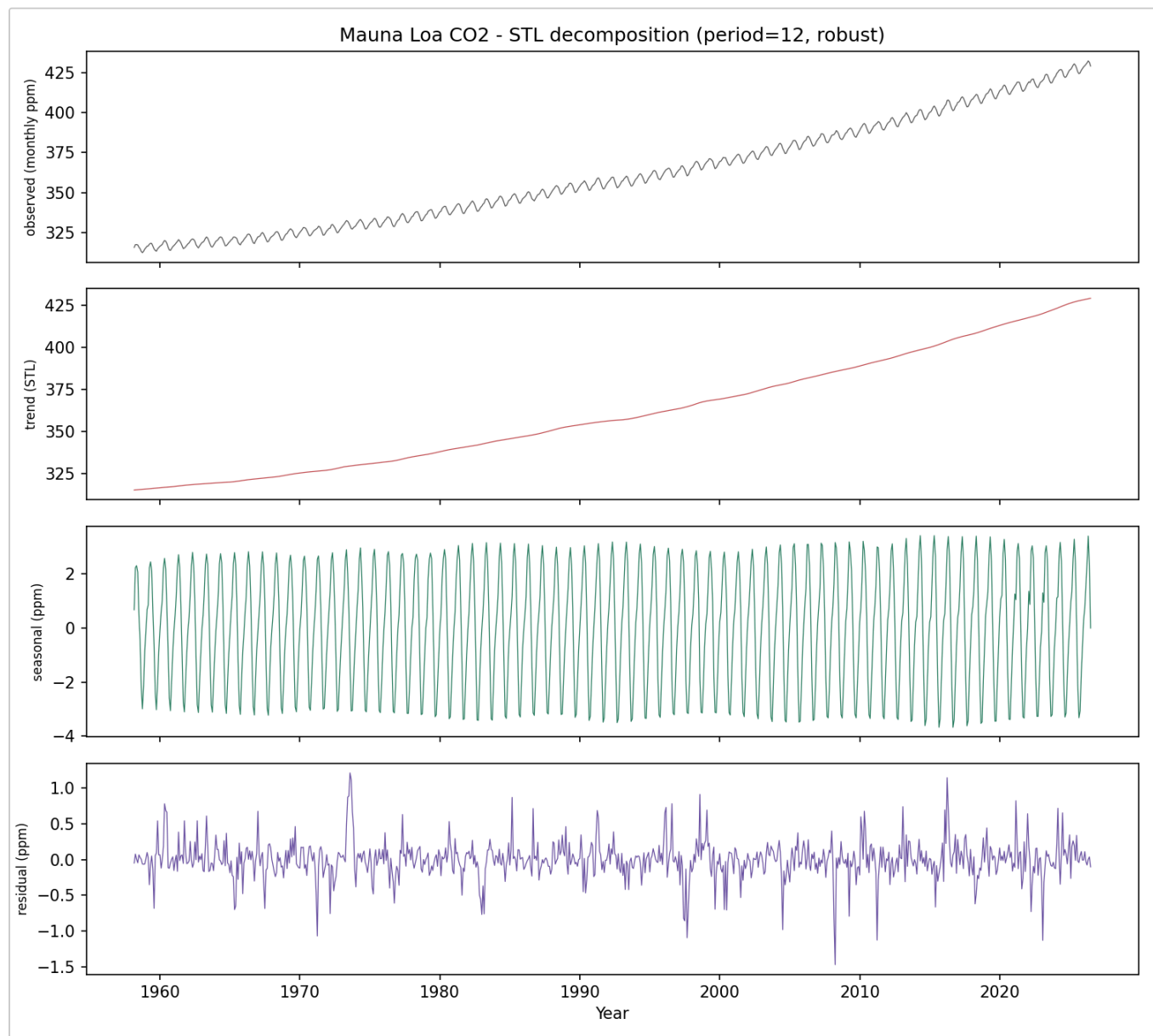


Figure 5. STL decomposition of monthly Mauna Loa CO2 into observed, trend, seasonal, and residual components.

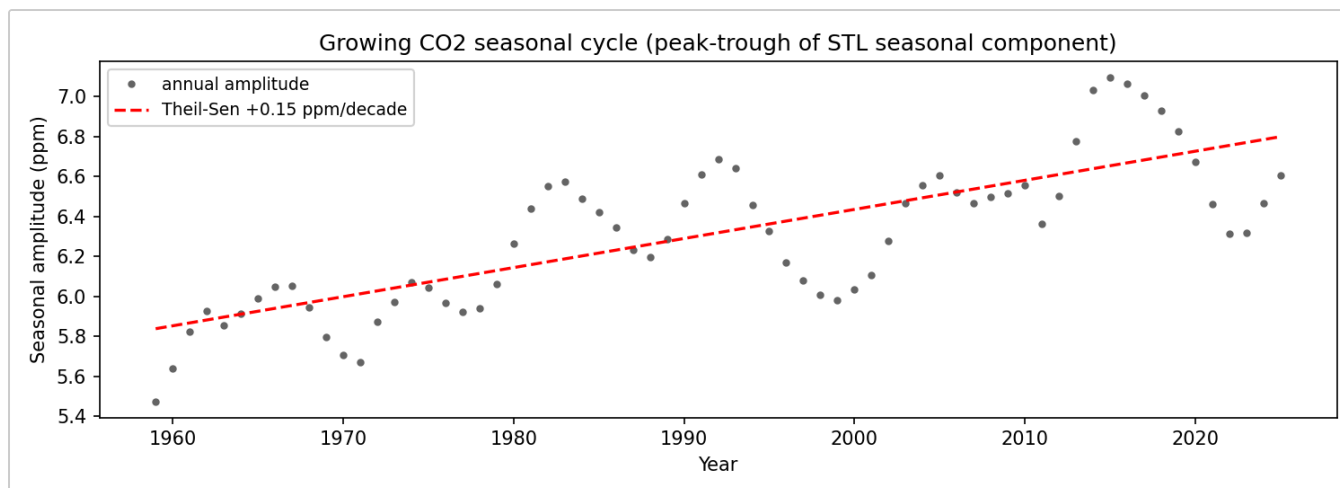


Figure 6. Annual peak-to-trough amplitude of the seasonal component (full calendar years, 1959–2025), with fitted trend.

Table 7. Growth of the seasonal cycle amplitude (full calendar years only) and residual diagnostics.

Quantity	Estimate	95% CI	p-value
Amplitude growth (ppm/decade, 67 full years)	+0.146	[+0.116, +0.179]	8.86e-12
Residual SD (ppm)	0.266	—	—

Figure 5 shows the four components of the decomposition. The trend component rises monotonically from approximately 315 to 430 ppm; the seasonal component is a regular oscillation whose amplitude increases over the record; and the residual is small ($\sigma = 0.266$ ppm against a signal of order 430 ppm) with no systematic structure, indicating that the additive trend-plus-seasonal model captures essentially all of the variance. The seasonal amplitude — the annual peak-to-trough difference of the seasonal component — increased by +0.146 ppm per decade (95% CI [+0.116, +0.179]; $p = 8.86e-12$; Table 7, Figure 6).

This growth has a natural physical reading. The seasonal amplitude is, in effect, the annual "breath" of the terrestrial biosphere: CO₂ falls through the northern summer as plants draw it down, and rises again through the winter as respiration and decomposition return it to the atmosphere. An amplitude that grows from year to year is consistent with an intensifying seasonal carbon exchange — consistent, for example, with longer growing seasons or CO₂ fertilisation of photosynthesis. The decomposition quantifies the change but does not, by itself, identify which of these mechanisms dominates; that would require additional data, such as satellite vegetation indices.

5.5 Periodicity of solar activity

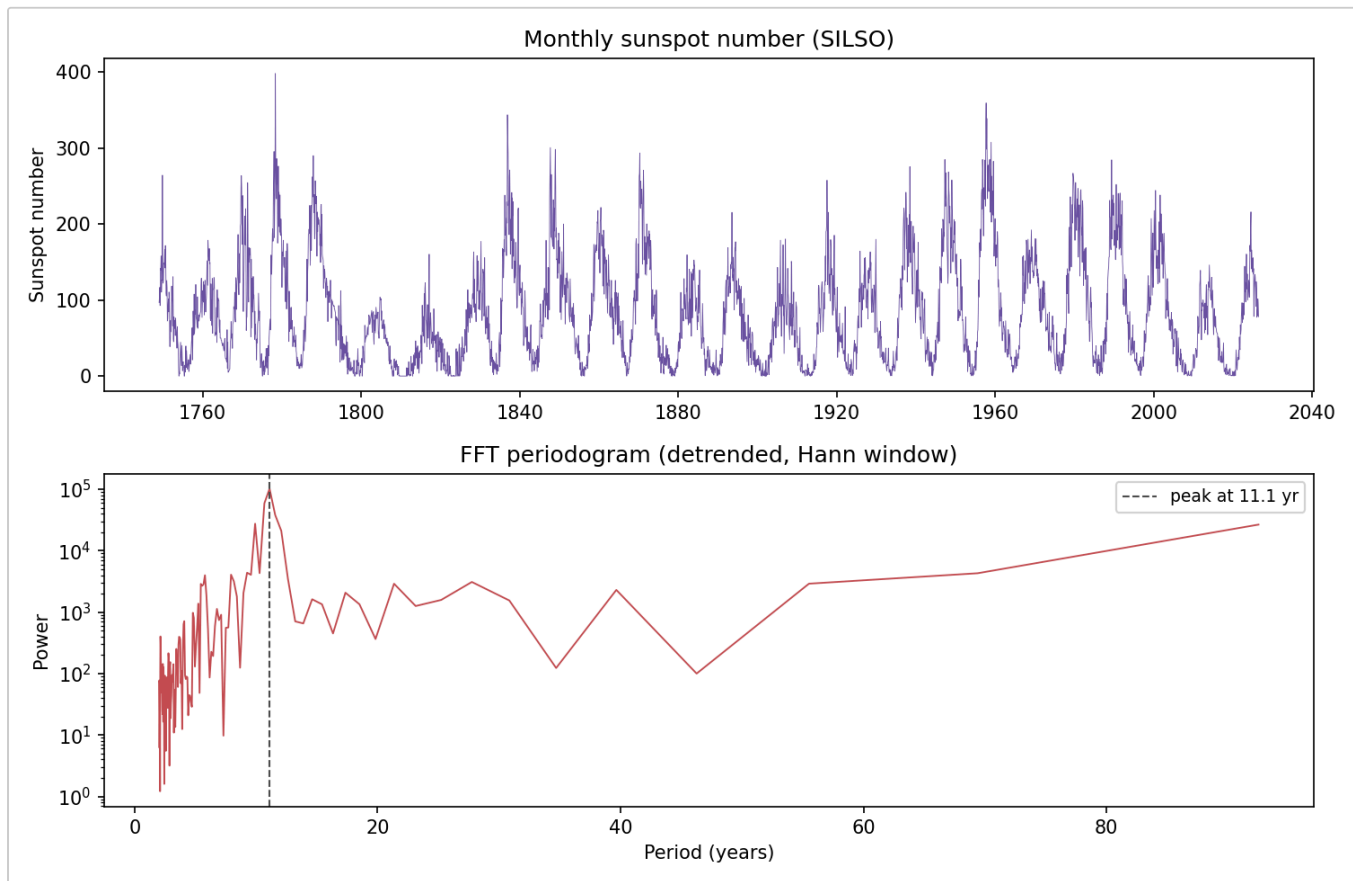


Figure 7. Monthly sunspot numbers (upper) and the FFT periodogram of the detrended series (lower), with the dominant peak marked.

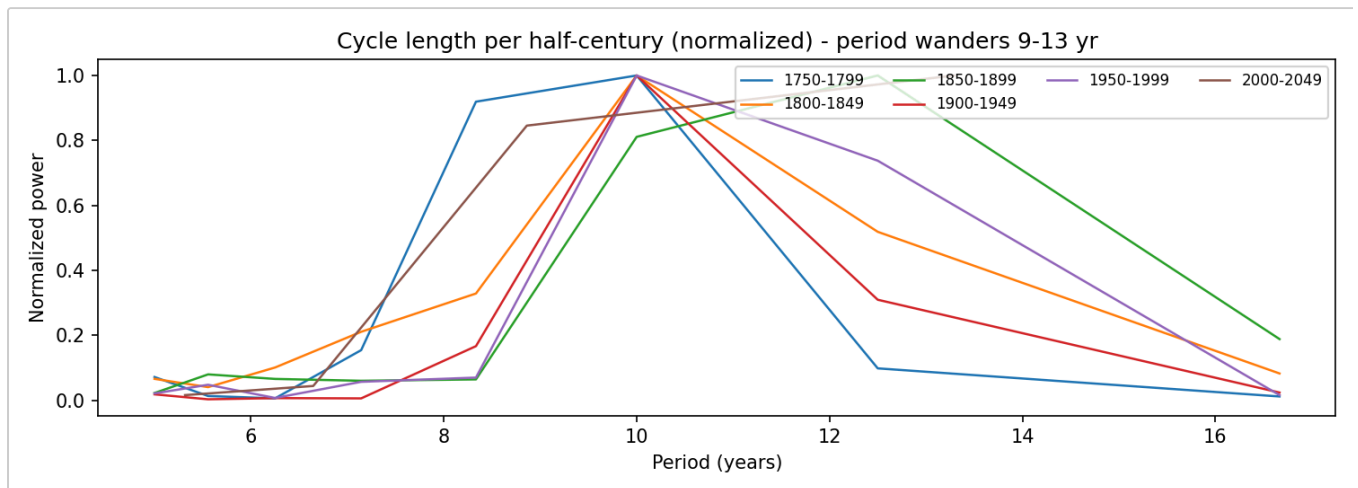


Figure 8. Normalised periodogram computed over successive half-century windows, showing the wandering of the peak.

Table 8. Dominant period of solar activity.

Quantity	Value
Dominant period (years)	11.10
Half-power span (years)	10.7–11.1

The periodogram of the detrended sunspot series exhibits a single dominant spectral peak at 11.10 years (Figure 7, lower panel), recovering the well-known Schwabe cycle from raw monthly counts without prior knowledge. The half-power span of the peak is 10.7–11.1 years (Table 8). Figure 8 shows that the peak location varies between approximately 9 and 13 years across successive half-century windows, reflecting the known irregularity of the solar cycle; the nominal 11-year period is therefore best read as a mean value rather than a strict constant.

5.6 Attribution: CO2 versus solar variability

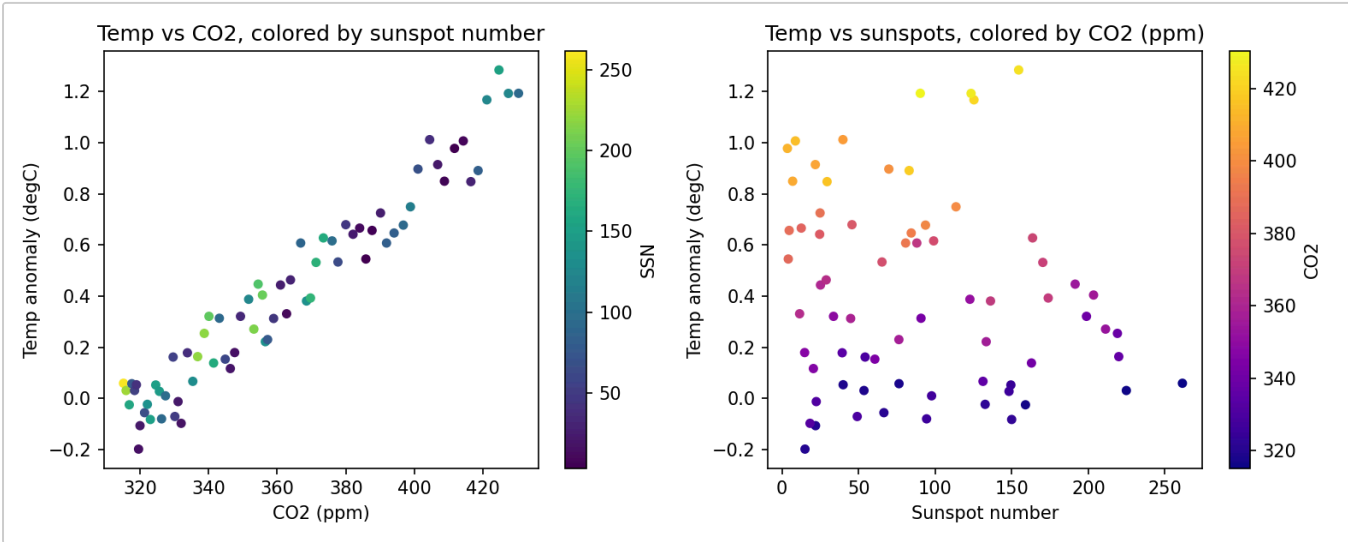


Figure 9. Left: temperature versus CO2, coloured by sunspot number. Right: temperature versus sunspots, coloured by CO2. Annual means, 1958–2026.

Table 9. Simple and partial Pearson correlations (n = 69 annual observations, 1958–2026). r^2 is the squared correlation: for simple correlations, the share of variance explained; for partial correlations, the share of residual variance explained after removing the control variable.

Correlation	r	r ²	p-value
Temperature – CO2 (simple)	+0.968	0.937	3.75e-42
Temperature – sunspots (simple)	-0.153	0.023	0.208
Temperature – CO2 sunspots (partial)	+0.972	0.945	1.02e-43
Temperature – sunspots CO2 (partial)	+0.351	0.123	0.00313

Table 9 reports the simple and partial correlations. Over the common period 1958–2026, the temperature–CO2 correlation is $r = +0.968$ ($r^2 = 0.94$; $p = 3.75e-42$) and remains essentially unchanged when the solar cycle is partialled out ($r = +0.972$; $r^2 = 0.94$; $p = 1.02e-43$): the partial CO2 association accounts for 94% of the temperature variance remaining after removal of the solar cycle. By contrast, the raw temperature–sunspot correlation is negligible ($r = -0.153$; $r^2 = 0.02$; $p = 0.208$), and although the partial correlation is statistically detectable it is small in comparison ($r = +0.351$; r^2

= 0.12): solar activity explains 12% of the residual temperature variance after removal of CO₂, roughly an eighth of the share associated with CO₂. These correlations are descriptive and do not by themselves establish causation; moreover, both temperature and CO₂ trend strongly over the common period, so the raw correlations partly reflect a shared trend (see the robustness check below). Figure 9 makes the asymmetry visible: temperature against CO₂ (left) falls on a tight line with no structure in the colour channel (sunspot number), whereas temperature against sunspots (right) shows no coherent pattern once the colour scale (CO₂) is accounted for.

Supplementary robustness check (confounding and autocorrelation). Removing a linear trend from both series reduces the temperature–CO₂ correlation to $r = 0.66$ ($p = 7 \times 10^{-10}$), and first-differencing reduces it further to $r = 0.31$ ($p = 0.009$); the association therefore survives both detrending and differencing and is not purely an artefact of the shared trend. The solar association does not survive differencing: the detrended temperature–sunspot correlation is $r = 0.30$ ($p = 0.013$), but the first-difference correlation is $r = 0.08$ ($p = 0.54$), so the solar signal is weak and sensitive to the detrending choice. Finally, all p-values in this section assume independent annual observations; the temperature residuals have lag-1 autocorrelation ≈ 0.5 (effective sample size ≈ 23 of 69 years) and the CO₂ residuals are near-unit autocorrelated, so the reported p-values are optimistic and should be read as upper bounds on significance.

5.7 Extreme years and documented climate events

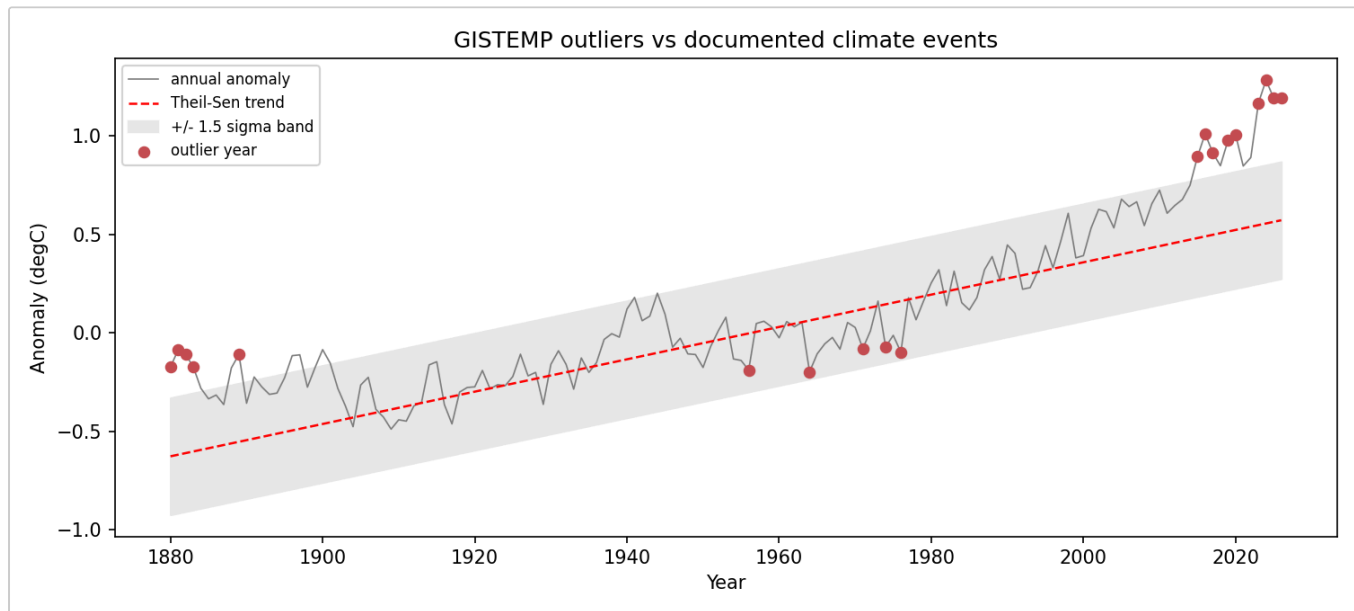


Figure 10. Annual temperature anomalies with fitted trend, the $\pm 1.5\sigma$ band of detrended values, and flagged outlier years.

Table 10. Years whose detrended anomalies exceed 1.5 standard deviations, with documented event matches (interpretation).

Year	z-score	Event match
1964	-1.92	Agung eruption (1963) cooling
1976	-1.90	1974 -1.69
1971	-1.61	1956 -1.54
2015	+1.51	2017 +1.51
1883	+1.58	1889 +1.65
1880	+1.70	2019 +1.75
2020	+1.86	1882 +1.94
2016	+2.05	El Nino 2015-16
1881	+2.10	2023 +2.54
2026	+2.55	2025 +2.59
2024	+3.09	El Nino 2023-24

Table 10 lists the years whose anomalies, after removal of the secular trend, exceed 1.5 standard deviations. Three flags correspond to documented events: 1964 (cooling following the 1963 Agung eruption), and 2016 and 2024 (El Niño years). The remaining flags fall in the 1880s, a period of lower data quality and higher relative noise. Equally informative are the documented events that do not reach the threshold, all of which admit a physical explanation: the 1992–93 Pinatubo cooling ($z = -0.95/-0.95$), muted by a concurrent El Niño; the 1998 El Niño ($z = +0.75$), whose anomaly the secular trend had overtaken; and 1983 ($z = -0.12$), whose El Niño warming was offset by the El Chichón eruption. The procedure therefore detects genuine event signals, and its apparent misses are attributable to documented physical offsets rather than to a failure of the detection method.

5.8 Forecasting comparison

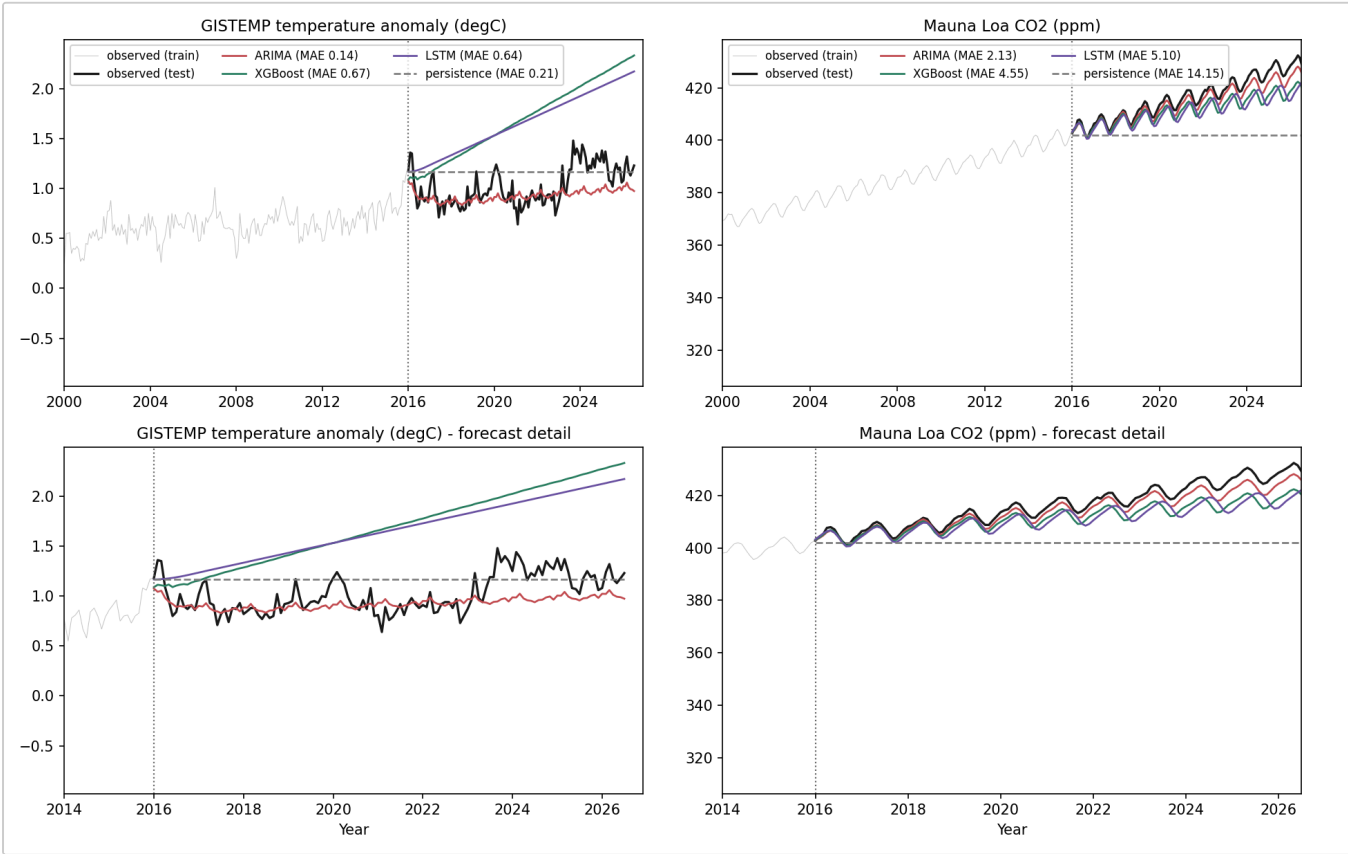


Figure 11. Forecasting comparison. Upper row: context since 2000. Lower row: forecast detail since 2014 — observed test values in black, model forecasts coloured, persistence dashed; the dotted line marks the train/test boundary.

Table 11. Holdout performance (test period 2016-01 to 2026-07, 127 months; MAE and RMSE, with improvement relative to the persistence baseline).

Series / model	MAE	RMSE	vs persistence
CO2 — persistence	14.153	—	—
CO2 — SARIMA	2.128	2.465	+85.0%
CO2 — XGBoost	4.550	5.472	+67.9%
CO2 — LSTM	5.096	6.234	+64.0%
Temperature — persistence	0.207	—	—
Temperature — SARIMA	0.139	0.183	+32.9%
Temperature — XGBoost	0.673	0.736	-225.1%
Temperature — LSTM	0.637	0.682	-207.7%

All models were fitted on data up to December 2015 and evaluated exclusively on the 127-month holdout 2016-01 to 2026-07 (Table 11, Figure 11); all rankings refer to this single holdout. For CO2, every method outperforms the persistence baseline by a substantial margin — SARIMA (MAE 2.128

ppm, an improvement of +85.0%), XGBoost (4.550 ppm, +67.9%), and LSTM (5.096 ppm, +64.0%) — consistent with the strong trend and seasonality of the series. For temperature, the ordering differs: only SARIMA improves on the baseline (MAE 0.139 °C, +32.9%), whereas both machine-learning models underperform it (XGBoost 0.673 °C; LSTM 0.637 °C). This contrast is consistent with the different character of the two series — CO₂ combines a strong trend with a pronounced seasonal cycle, whereas temperature behaves approximately as a random walk about a slow trend — but the experiment itself establishes only the measured error ranking. Model configurations were fixed in advance and are documented in `docs/process.md` §8.

Figure 11 also makes clear why the temperature forecasts do not track the test record closely. Two distinct effects are at work. The first is irreducible: the observed test values swing between roughly 0.6 and 1.5 °C from month to month, because monthly anomalies are dominated by short-term variability — weather and El Niño-scale fluctuations — that a model trained only on monthly lags cannot anticipate. SARIMA's forecast (MAE 0.139 °C) is a smooth conditional expectation: it is centred near the test level (its mean of 0.93 °C is close to the test mean of 1.02 °C) but, like any such expectation, it cannot follow the individual swings. The second effect is systematic: the machine-learning forecasts are built by compounding predicted monthly differences, and a small upward bias in those differences accumulates over the 127-month horizon. XGBoost's line therefore climbs to roughly 1 °C above the observed values in the final years (its MAE of 0.673 °C is nearly five times SARIMA's, and the LSTM's 0.637 °C is similar); the differencing introduced in Section 4 removed the gross drift of the original configuration, but the compounding mechanism remains. Neither effect can be removed by a lag-based model without external predictors.

6. Discussion and limitations

The principal caveat concerns the attribution analysis, which is descriptive rather than causal: partial correlation shows that the two candidate drivers separate cleanly in the data, but it cannot establish that CO₂ causes warming — it demonstrates consistency with that hypothesis and inconsistency with a dominant solar explanation. The breakpoint analysis rests on a single split at 1970, fixed by the project plan (which specifies the question as the warming rate before and after ~1970); because the plan and the results are contemporaneous in the project record, strict pre-specification of the exact year cannot be independently verified, although no evidence of data-driven selection was found.

The forecasting comparison rests on a single train/test split; a rolling-origin evaluation over multiple windows would provide more robust model rankings at greater computational cost. Model configurations were fixed rather than tuned — a tuned LSTM might perform better on CO₂, but tuning on the test period would invalidate the comparison.

All significance tests in this report (Mann–Kendall and Pearson) assume independent observations, whereas monthly and annual climate series are autocorrelated; the p-values are therefore optimistic.

Sections 5.2 and 5.6 report robustness checks with an effective-sample-size correction and with detrended and first-difference correlations, and the substantive conclusions are unchanged. The 2026 GISTEMP annual value uses only January–July, and the CO₂ seasonal-amplitude analysis is restricted to full calendar years; excluding the partial 2026 temperature year changes the full-record trend only in the third decimal place (0.082 → 0.081 °C per decade).

Finally, the data themselves impose limits. Recent GISTEMP values are preliminary and subject to revision, and the final months of the record are reported as missing and were excluded from the analysis.

7. Conclusions

- **Warming is real and accelerating.** The temperature trend is significant at any conventional level: 0.082 °C per decade over the full record and +0.204 °C per decade since 1970, with non-overlapping confidence intervals (Mann–Kendall $p \sim 10^{-39}$ under independent observations; $p \sim 10^{-4}$ after an effective-sample-size correction; Section 5.2).
- **The carbon cycle is intensifying.** CO₂ is rising at 16.422 ppm per decade and the amplitude of its seasonal cycle is growing (+0.146 ppm per decade), a change consistent with an intensifying seasonal carbon exchange.
- **Solar variability is unlikely to be the dominant explanation for the warming.** The solar cycle is a well-defined 11.10-year oscillation, yet the temperature–CO₂ association survives removal of the solar cycle (partial $r = +0.97$) while the residual solar association after controlling for CO₂ is small (partial $r = +0.35$; $r^2 = 0.12$), does not survive first-differencing, and is not robust given the limited effective sample size.
- **SARIMA produced the lowest forecast error.** Under the fixed configurations and the single 127-month holdout, SARIMA outperformed both XGBoost and a small LSTM on both series: for CO₂, XGBoost and LSTM beat persistence but not SARIMA; for temperature, they performed substantially worse than persistence. Rankings are conditional on this holdout and configuration set.

Taken together, the three records tell a coherent story. The warming of the past century is real, statistically unambiguous, and accelerating; it is strongly associated with the rise of greenhouse gases rather than with solar variability; and the seasonal amplitude of atmospheric CO₂ is growing. The forecasting results add a methodological coda: under this holdout configuration, the classical seasonal model produced the lowest forecast error on both series.

8. Reproducibility

Every result is generated by a single command:

```
python experiments/report.py      # reruns all stages; regenerates plots and this report
python experiments/<stage>.py     # or any single stage in isolation
```

Outputs comprise this report (`results/report.html` and `results/report.pdf`), the plots in `results/plots/`, and the living log `docs/process.md`, which records each stage's setup, numbers, and reasoning. Raw data in `data/raw/` are never modified by analysis code.